

Linking East Asian Dust Flux to Surface Ocean Carbonate Chemistry in the Kuroshio-Oyashio Extension: A 23-kyr Record of Clay Mineralogy and Boron Isotopes

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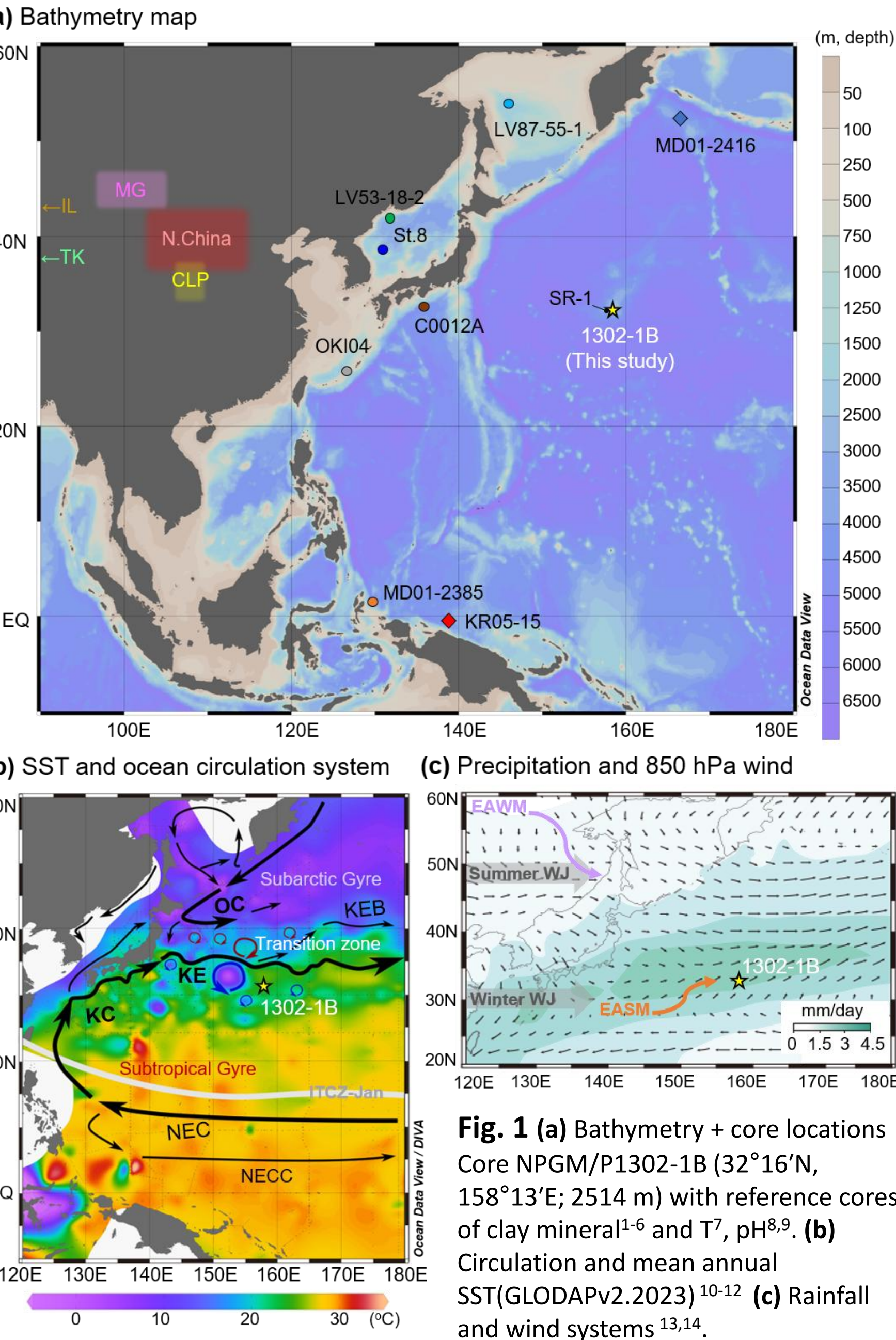
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Background & Aim

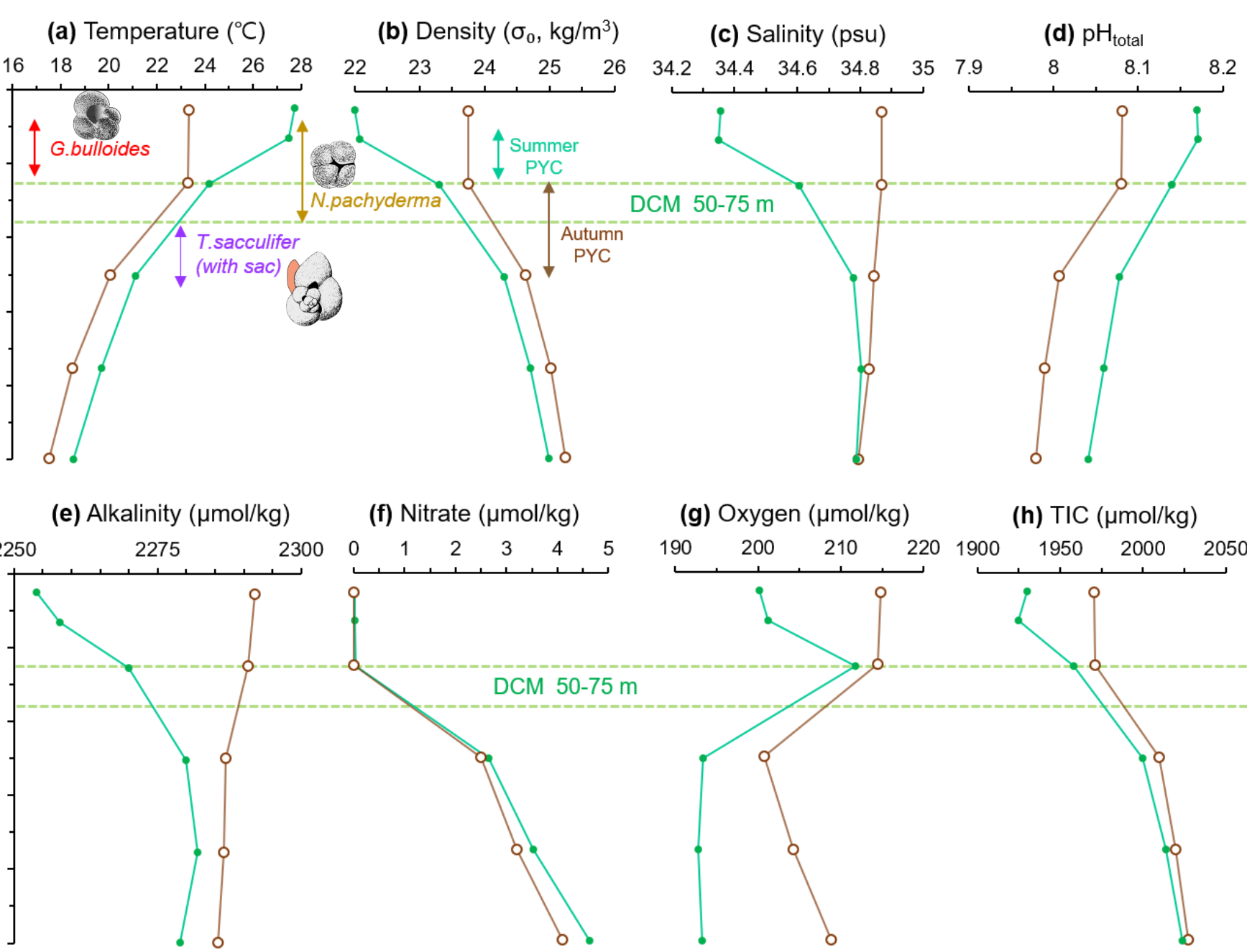
- NW.Pacific, KE (Kuroshio-oyashio Extension) area: subtropical/subpolar gyre boundary

In this study:

- Atmospheric changes: Eolian provenance via clay mineral composition + magnetic properties
- Oceanographic changes: Multi-depth pH from foraminiferal $\delta^{11}\text{B}$ (3 species)
- Understanding the atmospheric + oceanographic forcing during the last deglaciation (~23 ka)



Foraminifera species thriving depth
@ Mixed layer: *G. bulloides* (~50 m), *N. pachyderma* (~20-75 m)
@ Below DCM: *T. sacculifer* with sac (> 75 m)
The DCM is estimated at ~50–75 m depth in study area



Methods

Age model

¹⁴C AMS dating of mixed planktic foraminifera (Beta Analytic, USA), Cal BP using Calib v7.1; error < 70 yr, reservoir age 414 yr¹⁷

Boron isotope & trace element analysis

¹¹B of *G. bulloides* (300–355 μm), *N. pachyderma* (250–300 μm), and *T. sacculifer* with sac (425–800 μm) (MC-ICP-MS: NuPlasma3, SNU IRF) after column separation¹⁸; trace elements (Al, Ca, Mn, Sr) by ICP-MS (Elan DRC-II, KBSI)

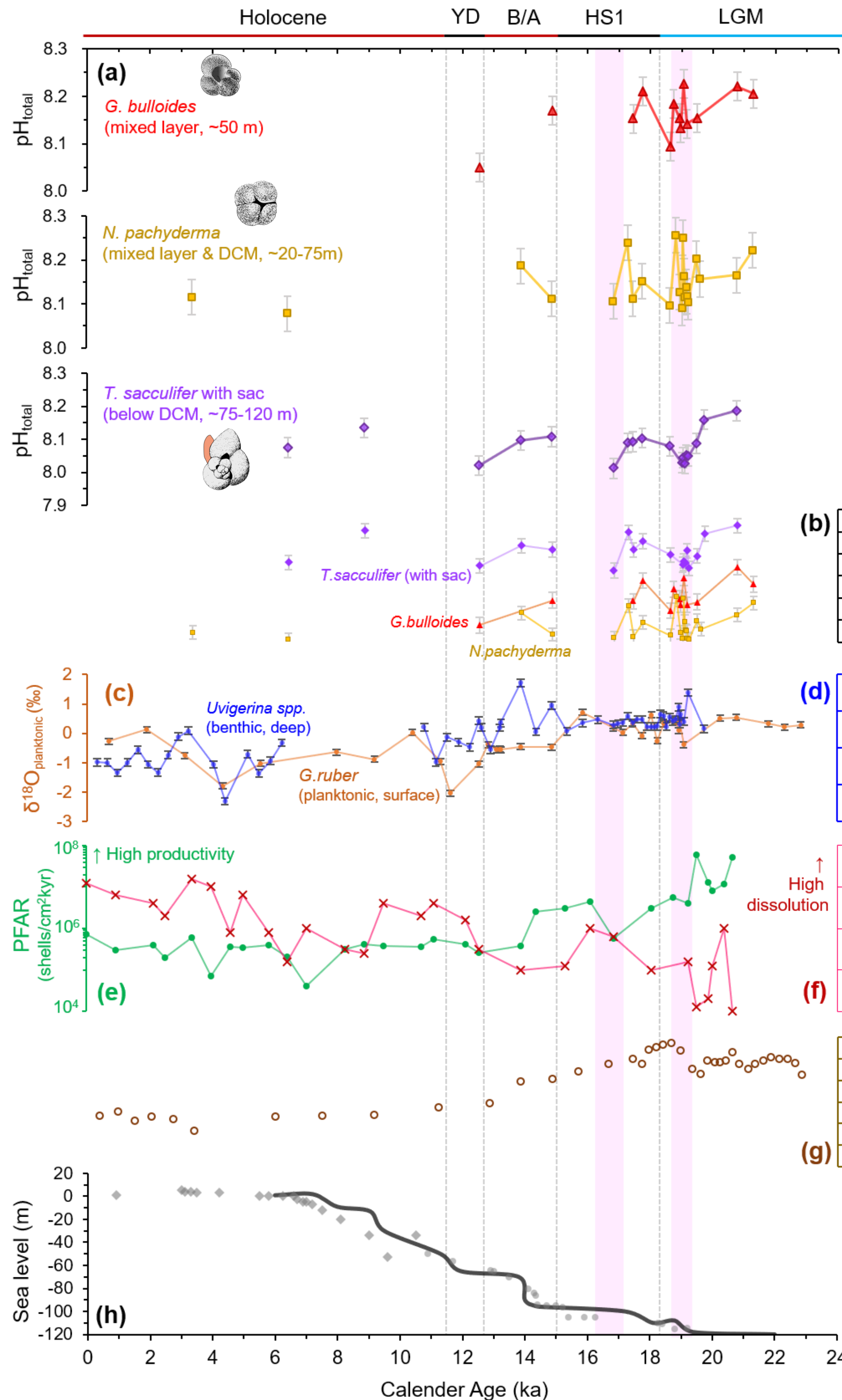
Clay mineral analysis

•XRD of <2 μm oriented samples (air-dried & ethylene glycol-saturated); Bruker D8 Advance A25 (GNU), Cu Kα (λ = 1.541 Å); obtain Illite crystallinity (IC)–FWHM of illite 10 Å peak, Illite chemical index (ICI) –5 Å to 10 Å peak area ratio and proportions of illite, smectite, kaolinite, chlorite by weighting factors¹⁹

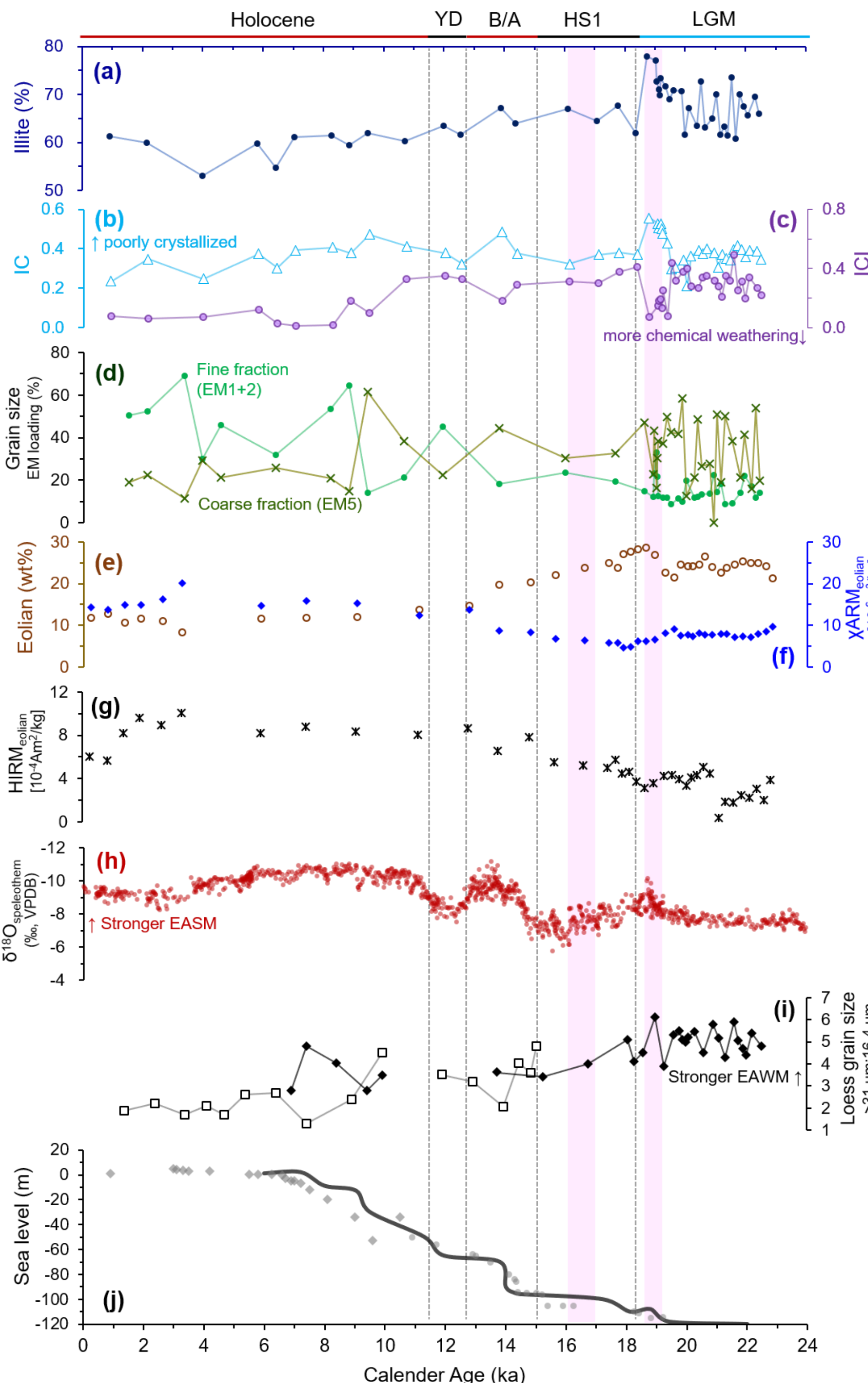
Grain-size end-member modelling (EMMA)

•Laser diffraction particle size analysis (Beckman Coulter LS 13 320 XR); 5-end-member model fitted using AnalySize v1.1²⁰

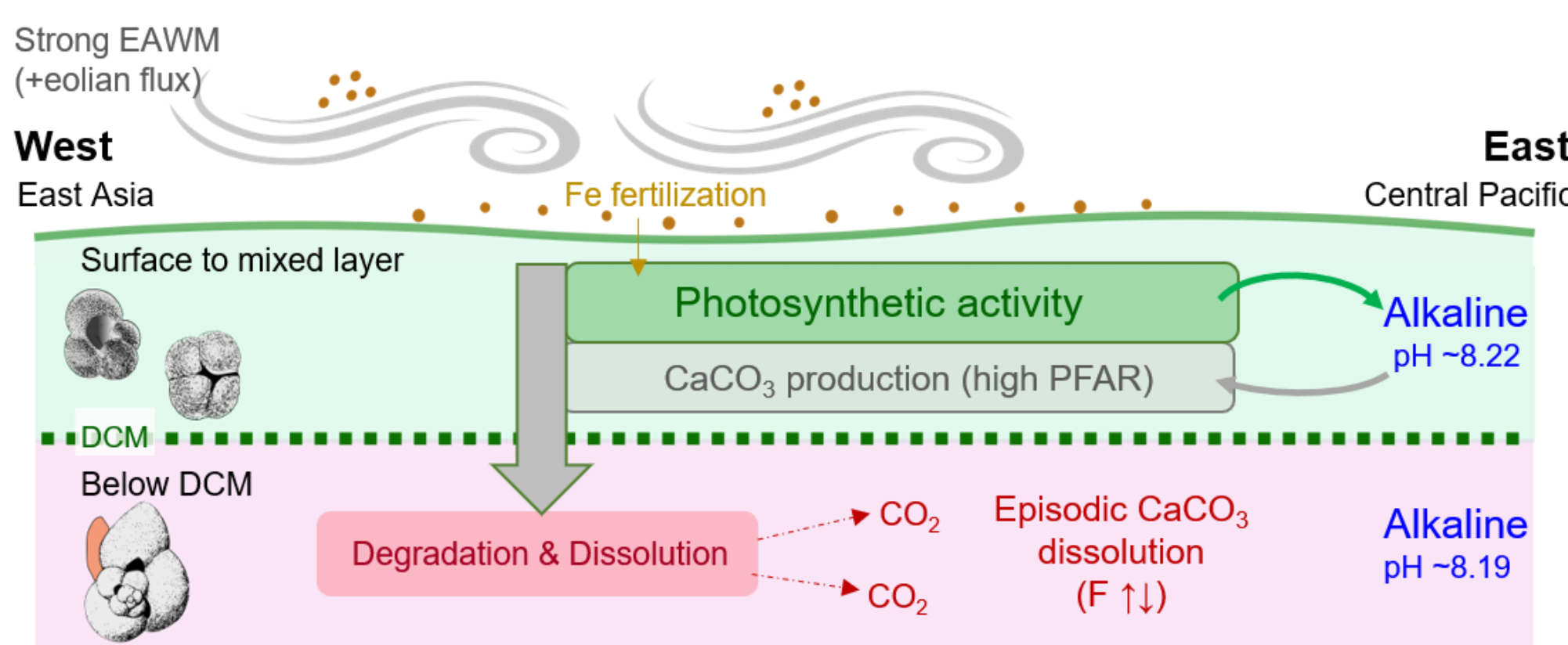
Depth-dependent pH divergence peaks at LGM termination



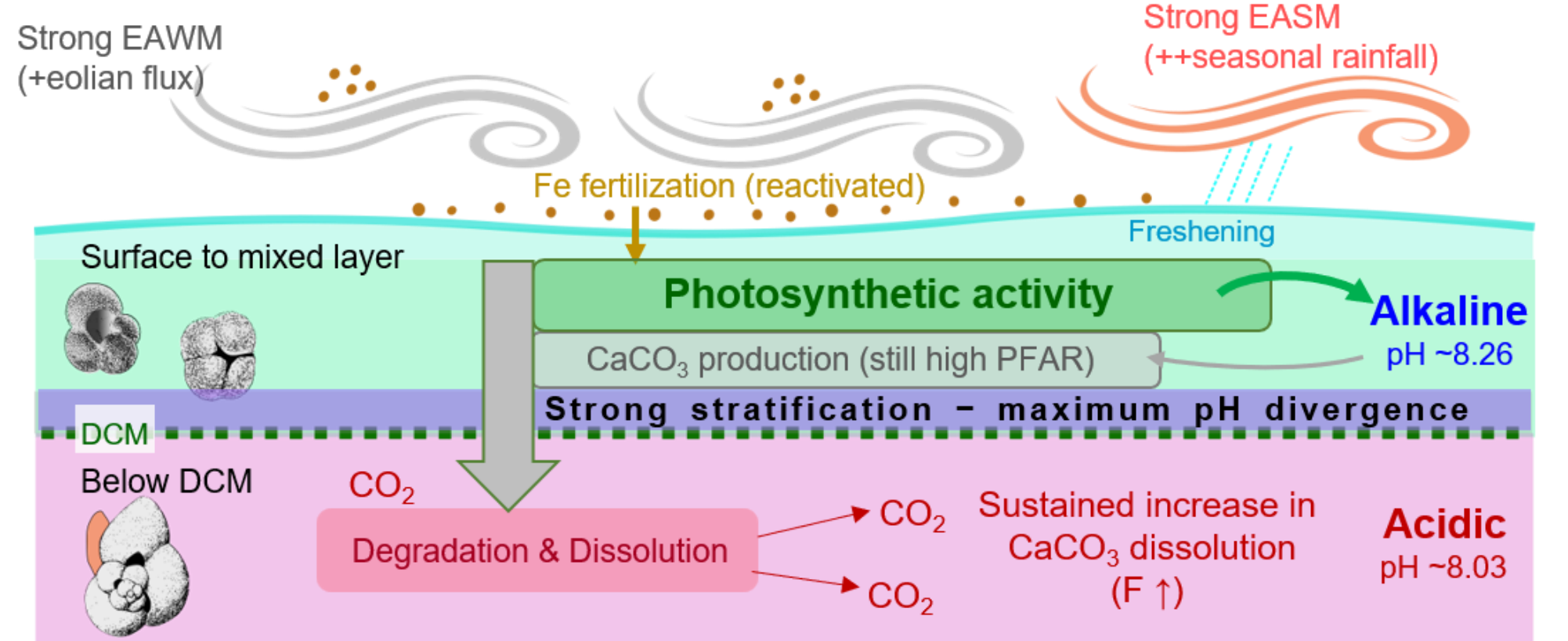
Eolian surge ← monsoon co-intensification, exposed continental shelves



(a) Late LGM (~21.3–19.4 ka)



(b) LGM termination (19.3–18.7 ka)



(c) Mid-HS1 (~16.8 ka)

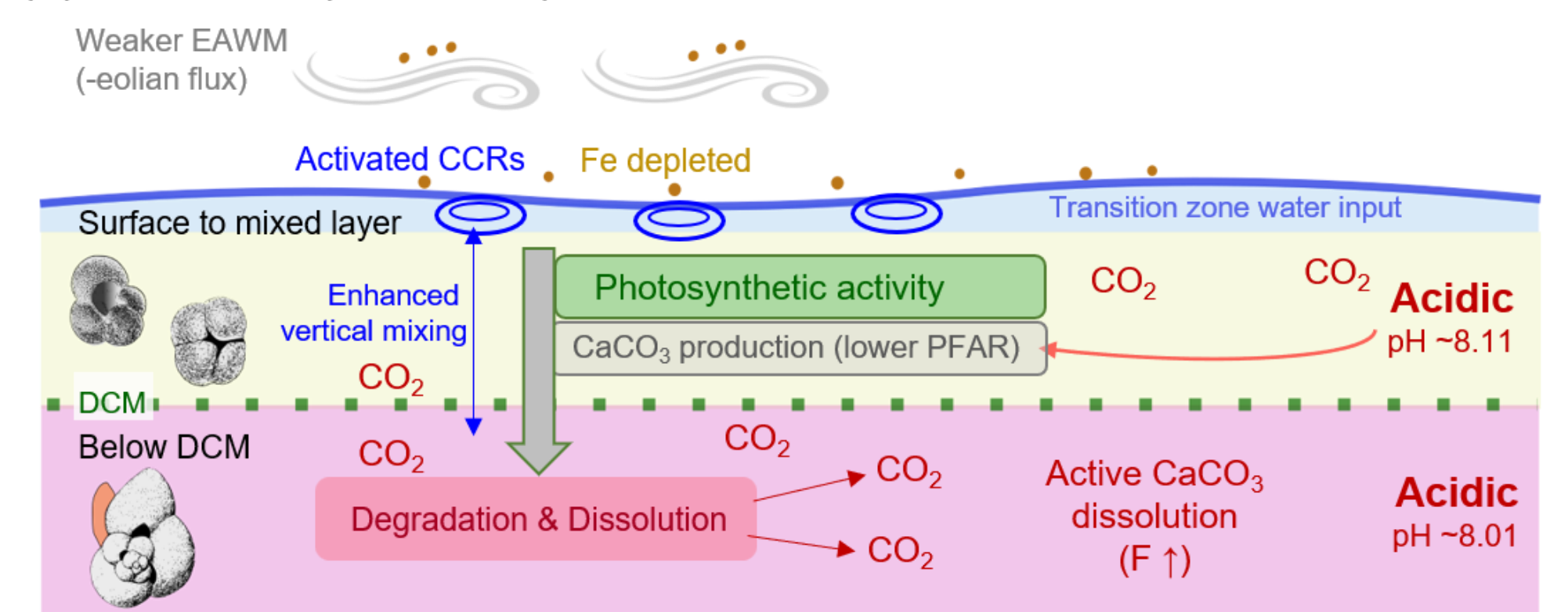


Fig. 4 (a-c) Schematic pH change mechanisms: late LGM, LGM termination, mid-HS1. Arrow thickness denotes relative intensity of eolian flux and organic matter export.

Strong EASM + EAWM → High eolian flux + seasonal rainfall
→ Fe fertilization by dust + stratification in seawater
→ depth-dependent pH divergence at LGM termination

High illite % = dominant Asian dust provenance

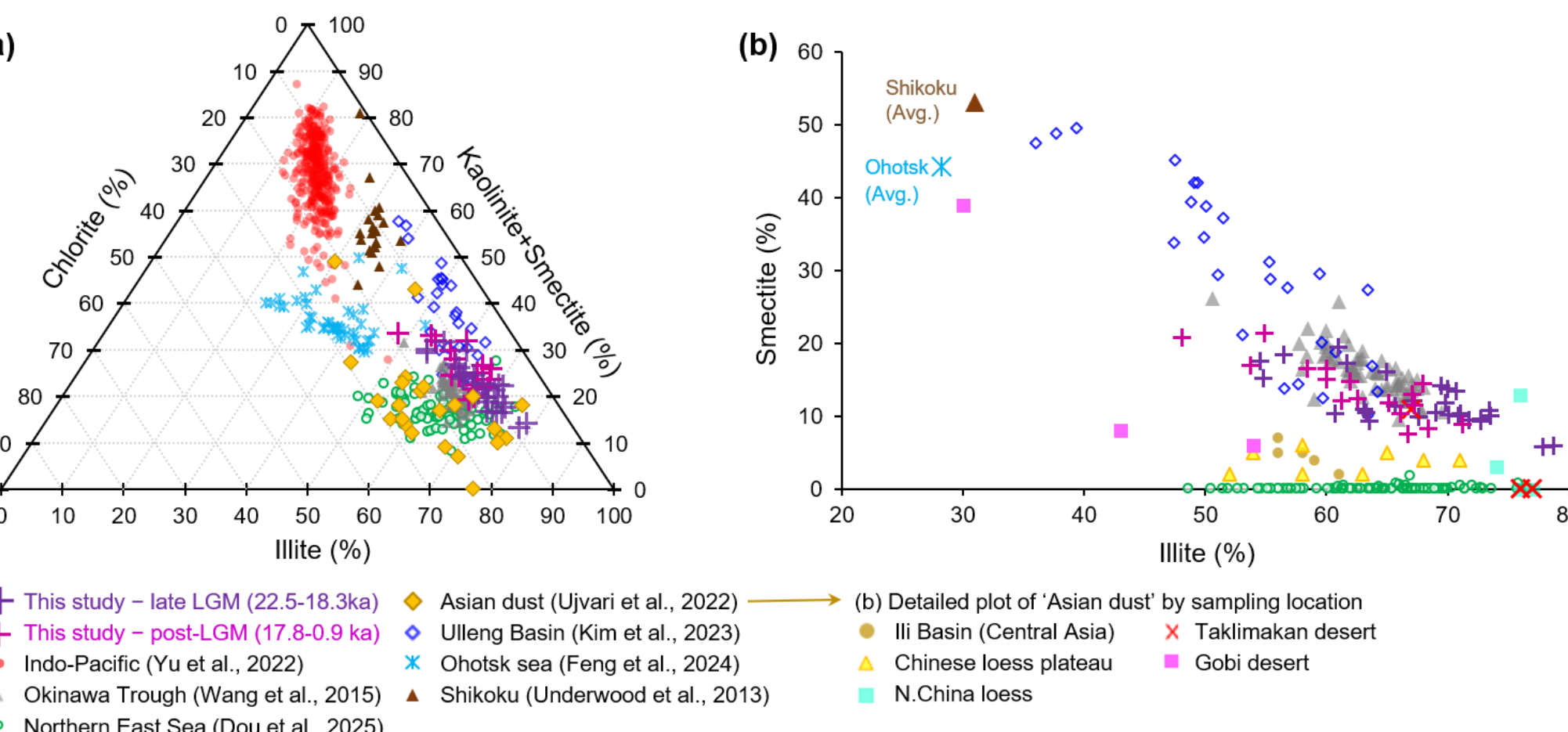


Fig. 6 Clay mineral composition of the study core and potential source regions^{1-6,29}
(a) Ternary diagram of smectite-illite-(chlorite+kaolinite), (b) Cross plot of illite vs smectite

pH divergence across western Pacific

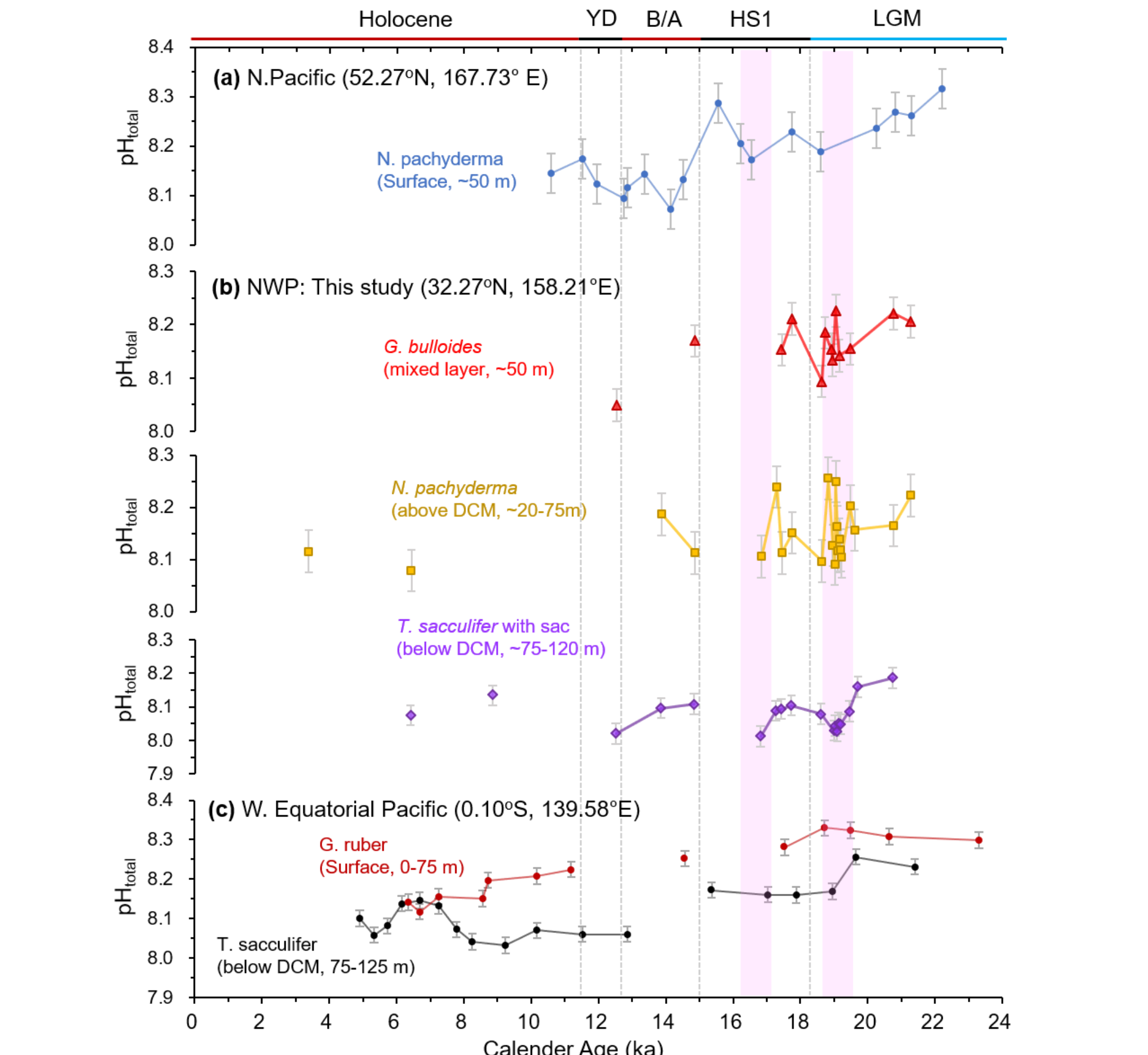


Fig. 7 pH records from (a) N. Pacific⁸, (b) NWP (this study), (c) W. equatorial Pacific⁹